

Hossein Rezaei

Applied Chemistry Student

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ACADEMIC PROFILE

Undergraduate student pursuing a B.Sc. in Applied Chemistry at Khatam al-Anbia University of Behbahan, currently holding a GPA of 17.9/20. Academic interests center on catalysis and green chemistry, supported by developing laboratory training and instrumental analysis exposure. Preparing academically and linguistically for graduate study (Master's) in Europe.

EDUCATION

B.Sc. in Applied Chemistry

Khatam al-Anbia University of Behbahan, Behbahan, Iran

2022 – Present | GPA: 17.9 / 20

Coursework in organic, inorganic, physical, and analytical chemistry with laboratory training in instrumental analysis and separation techniques.

- Internship: Bid Boland Gas Refinery (Summer Internship)
- Secretary of the Foreign Languages Association at the university

SCIENTIFIC INTERESTS

- Catalysis (Organometallic & Homogeneous Catalysis) — design and development of metal-based catalysts for selective transformations
- Green & Sustainable Chemistry — development of environmentally benign chemical processes, reducing hazardous reagents and energy demand
- Organic Chemistry — reaction mechanisms and synthetic pathways
- Analytical Chemistry — instrumental methods for identification and quantification

INSTRUMENTAL ANALYSIS TRAINING

- Gas Chromatography (GC) — laboratory coursework exposure (9-hour workshop)
- High Performance Liquid Chromatography (HPLC) — supervised training sessions (included in 14-hour workshop)
- Atomic Spectroscopy (AAS) — introductory laboratory training (10-hour workshop)
- UV-Visible Spectroscopy — hands-on laboratory practice
- X-Ray Diffraction (XRD) — conceptual / demonstrative exposure (3-hour workshop)

TRAINING & WORKSHOPS

- Organometallic Catalysis in Sustainable Chemistry (DTU) — 10 hours
- Solution Preparation & HPLC-GC Separation Methods — 14 hours
- HSE Officer in Oil Projects (Skill Certificate) — 130 hours
- Atomic Spectroscopy Workshop — 10 hours

- X-Ray Diffraction Workshop — 3 hours
- GC Workshop — 9 hours
- ISI Scientific Article Writing Workshop — 6 hours
- Windows 11 Fundamentals — 6 hours

LABORATORY & DIGITAL SKILLS

- Solution Preparation — molarity calculations, standard/stock solutions, dilution
- Separation Techniques — filtration, extraction, distillation, chromatography basics
- Laboratory Safety Principles — chemical handling, PPE, SDS interpretation
- Digital Skills — Microsoft Office, basic data analysis/charting, ChemDraw (4 hours), Mendeley Reference Management

INDUSTRIAL EXPERIENCE

Chemistry Internship Trainee

Bid Boland Gas Refinery, Khuzestan, Iran

Summer Internship

Completed a supervised internship placement introducing laboratory and process-monitoring practices in an industrial setting. Involvement was observational and supportive under the guidance of refinery personnel, without independent operational or analytical responsibility. Followed refinery HSE protocols throughout the placement, including PPE use, hazard identification, and emergency response procedures.

INTERNATIONAL ACADEMIC PREPARATION

- Actively preparing for the TOEFL iBT exam, currently at B2 level, targeting a score of 5.5 in the new TOEFL iBT scoring system. (equivalent to approximately 107–113 on the previous scale)
- Familiarizing with European grading systems, application procedures, and program structures.
- Developing skills in academic writing, including CV preparation and statement-of-purpose composition.
- RELEVANT COURSEWORK

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- General & Inorganic Chemistry
 - Organic Chemistry I & II
 - Physical Chemistry
 - Analytical Chemistry
 - Instrumental Analysis Laboratory
 - Industrial Chemistry
 - Chemical Safety & Laboratory Ethics